

Fig. 2: Data sampler

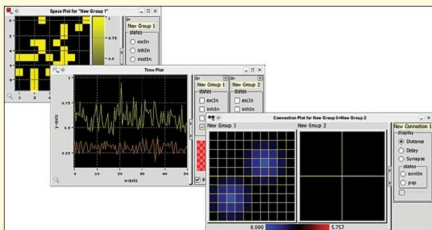


Fig. 3: Space, time and connection plots

in a format based on XML standard, and can thus be transformed to a wide range of other description grammars for neuronal models.

Predefined interfaces to hardware devices. There is a wide range of predefined interfaces to hardware devices in iqr. Modules can run synchronised or in their own thread, independent of the updated speed of the main simulation. The main predefined interfaces include modules to control Khepera and Koala robots, Lego MindStorms and blimp robots.

Changing model parameters at run-time. The possibility of changing model parameters at run-time helps the designer analyse responses of the system in various test cases and allows him or her to come into conclusion about the behaviour of the system.

Projects using iqr

A large number of projects have made use of this excellent tool for neural simulation. iqr was used for building distributed adaptive control (DAC) series of models. In this project, a mobile robot could form associations about colour patterns during the exploration of an arena. Robots can precisely travel to target locations with the help of memory.

We can also make use of iqr to prepare models of classical conditioning. For example, a model of learning in the auditory cortex can be made with the help of this tool.

Competing neural simulators

Some competing neural simulators are:

NEURON. This is a simulation environment for modelling individual neurons and neuron networks. The latest version of NEURON is capable of handling diffusion-reaction models and integrating diffusion functions into models of synapses and cellular networks. The higher level of abstraction given by iqr, than highly-specific neuronal system simulators like NEURON, helps iqr to have its size reduced and speed increased.

General Neural Simulation System (GENESIS). This is yet another simulation environment for constructing realistic models of neurobiological systems. Functionalities of GENESIS are similar to that of NEURON.

NEURONC. This tool was originally developed for modelling nerve cells in the vertebrate retina. Many enhancements have been made on this tool over years. It is now used as a general-purpose simulator for modelling a large number of nerve cells, each with a large number of compartments. The neuron model used in iqr is a point-model with no compartments.

BRIAN. It is an easy-to-learn and use, highly-flexible and easily-extensible neural simulator intended not only to save time of processors but also scientists. The tool is written in Python and is available for almost all platforms.

MATLAB. iqr helps us have an added number of biological constraints than a general-purpose simulation tool like MATLAB. The domain-specific predefined blocks help in creating complex systems, more easily than MATLAB. Also, the biological language used in iqr makes it preferable than a general-purpose tool.

Synthetic insect. One interesting project using iqr is a synthetic insect. The model integrated a number of subsystems such as decoding celestial information into a heading direction signal, efficient and biologically-grounded storage of allo-centric information on distance and direction travelled, decoding of memory and computation of home vector, stabilisation of the course based on optic flow, prioritisation of actions and behaviour regulation.

The behavioural task assigned to this robot was to explore the environment while searching for the target stimulus and return to the point of departure upon either finding the target stimulus or while exceeding a given duration threshold of the exploration phase. The current under-

standing of the neuronal organisation of the foraging behaviour in insects is incorporated in the model.

The system that was developed was large-scale, comprising 132 groups of neurons organised into 30 processes. The organisation of this model is supported by iqr.

iqr is a very useful simulation environment for large-scale neuronal systems. The software is an outcome of extensive research being done to sort out the challenges faced by engineers, while trying to understand neuronal systems, especially for constructing artificial brain-like systems. If you are looking for a really simple and easy-to-use neural simulator, iqr could be the ideal one for you. ●

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